A UNIVERSALLY INJECTIVE ETALE MORPHISM IS AN OPEN IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author gives a second proof of the same lemma immediately after the counted span. The counted span delivers the first proof in full, headed as such by the author; the second proof lies outside the counted span and was not moved into it or summarised.

Editorial note. The proof carries the author's own heading '[First proof]'; it is reproduced verbatim.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Etale morphisms are open. Morphisms, Lemma morphisms-lemma-etale-open, invoked at proof lines 6627-6628. Lets the target be replaced by the image, so that the morphism may be assumed surjective.
- A flat, surjective, quasi-compact monomorphism is an isomorphism. Same chapter, Lemma lemma-flat-surjective-quasi-compact-monomorphism-isomorphism, invoked at proof lines 6631-6633. Reduces the remaining task to showing that the morphism is a monomorphism.
- The diagonal of an etale morphism is an open immersion, and universal injectivity makes it surjective. Morphisms, Lemma morphisms-lemma-diagonal-unramified-morphism at proof lines 6633-6635, Morphisms, Lemma morphisms-lemma-flat-unramified-etale at proof lines 6636-6637, Morphisms, Lemma morphisms-lemma-universally-injective at proof lines 6638-6639. Identifies the diagonal as an open immersion and then as surjective, so that it is an isomorphism, which is exactly the statement that the morphism is a monomorphism.

Source reference. Original lemma 06NC: sources/stacks-descent/pinned-original-source.tex, source lines 6599–6599. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02GE: sources/stacks-chow/tags.txt, source lines 3225–3225. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03WT: sources/stacks-chow/tags.txt, source lines 5124–5124. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02GV: sources/stacks-chow/tags.txt, source lines 3242–3242. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01S4: sources/stacks-chow/tags.txt, source lines 2325–2325. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *A universally injective étale morphism is an open immersion.*

First proof. Let $f : X \to Y$ be an étale morphism which is universally injective. Then f is open (Morphisms, Lemma 03WT) hence we can replace Y by $f(X)$ and we may assume that f is surjective. Then f is bijective and open hence a homeomorphism. Hence f is quasi-compact. Thus by Lemma 06NC it suffices to show that f is a monomorphism. As $X \to Y$ is étale the morphism $\Delta_{X/Y} : X \to X \times_Y X$ is an open immersion by Morphisms, Lemma 02GE (and Morphisms, Lemma 02GV). As f is universally injective $\Delta_{X/Y}$ is also surjective, see Morphisms, Lemma 01S4. Hence $\Delta_{X/Y}$ is an isomorphism, i.e., $X \to Y$ is a monomorphism. $\square$